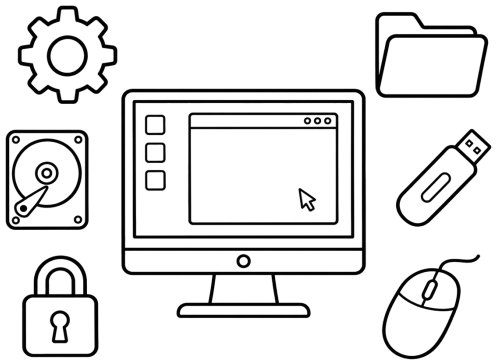
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 10



UNIT 5
Operating Systems

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain hardware, firmware, software, liveware and the booting process.
- [ ] Sort software into system software and application software.
- [ ] Describe the types of operating systems and their main services.
- [ ] Compare the command line interface (CLI) with the GUI.
- [ ] Explain how the OS manages processes, memory, devices and security.
- [ ] Work with file names, extensions, properties and file systems.
- [ ] Explain utility software, drives, partitioning and formatting.

THREE RULES FOR THE COMPUTER ROOM

- **Always** ask your teacher before you use Command Prompt or Task Manager.
- **Never** partition, format or defragment a drive on your own.
- **Always** keep a backup copy of your files, saved with your name.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical task on a computer. Last you **write** the exercises in this book.

You will need: a pencil and an eraser, your textbook (pages 138 - 157), a computer with Windows (or Linux), and a pen drive for the practical tasks.

THE WORDS YOU WILL LEARN

Hardware	Firmware	Software	Liveware
BIOS	POST	Bootstrap Loader	Booting
System software	Utility software	Language translator	Operating system
Multi-tasking OS	Real time OS	CLI and GUI	WIMP
Process	Device driver	File extension	File system
Random access	Partitioning	Formatting	Defragmentation

"Know your operating system, and your computer will work for you."

- THE KING OF ICT

CHAPTER 1

The Computer System and Software

? WHO STARTS THE COMPUTER?

"When I switch on our computer, some white text comes first. What is it doing?"

"It is booting! Then the operating system is loaded."

The most important software of a computer is its **operating system**.

5.1 Hardware, firmware, software and liveware

A computer consists of **hardware**, **firmware**, **software** and **liveware**.

Part	What it is	Examples
Hardware	Any physical component of a computer system that can be touched.	Mouse, keyboard, display unit, hard disk, speaker, printer
Firmware	The booting instructions stored in the ROM (Read Only Memory). The first text on the screen when the computer starts is shown by firmware.	The BIOS in the ROM
Software	A set of instructions given to the computer to perform some activity.	Operating system, word processing, web browsers
Liveware	The users who operate the computer.	You and your teacher at the computer

5.2 The booting process

When the computer is switched on, four things happen. This process is called **booting**.

1 POWER ON — Electrical power is supplied to the processor, RAM and ROM. →

2 POST — The BIOS stored in ROM performs the POST (Power-On Self-Test). →

3 BOOTSTRAP LOADER — The BIOS loads and executes the Bootstrap Loader from the boot sector. →

4 OS LOADED — The Bootstrap Loader loads the OS into RAM. The OS gets full control.

The BIOS (Basic Input Output System, textbook page 51) is stored in ROM. In the POST it configures the computer settings: the amount of RAM, the hard disk, monitor, keyboard, mouse and the boot sequence. The Bootstrap Loader is in the **boot sector** of the secondary storage.

"A computer needs hardware, firmware and software. It also needs you - the liveware."

- THE KING OF ICT



Colour me!

5.3 Types of software

Software is a set of instructions given to the computer to perform some activity. It is broadly classified into **system software** and **application software**.

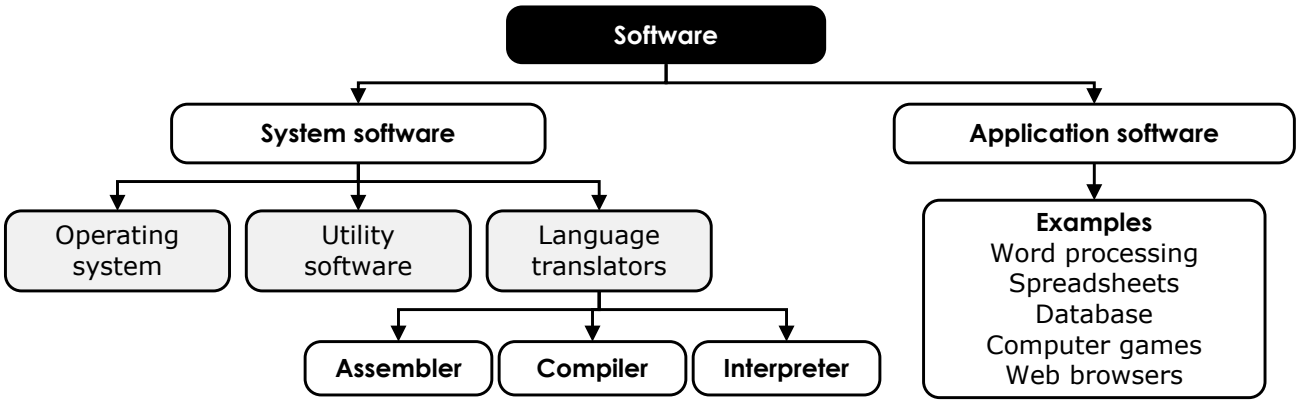


Figure 5.1 - types of software

A computer program is made of instruction codes, written in **high-level languages** that are very close to human languages. **Language translators** translate them into **machine language** (0s and 1s), which the computer understands: **assembler**, **compiler** and **interpreter**.

Application software runs on the operating system. It is used for computer-based activities of the user: creating documents, mathematical functions, data entry and computer games.

5.4 The operating system

The **operating system** (OS) lets the user use the functions of a computer by managing the hardware and the software in it.

- It gives the instructions to install and run application software.
- It manages the input devices, the output devices and the memory.
- It handles the other software installed on the computer.

So the OS carries out the overall operation of the computer. Figure 5.2 shows how the software works with the hardware.

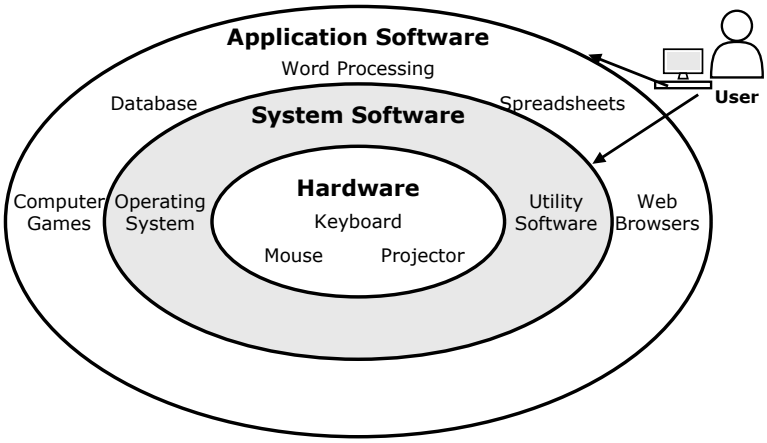


Figure 5.2 - the hardware and the software

"Every morning you boot up too. Start the day with a good self-test!"

- THE KING OF ICT

5.5 Several operating systems

Some operating systems are **proprietary** and some are **free and open source** (page 141).

Operating system	Free and open source / Proprietary	Usage
Microsoft Windows	Proprietary	Desktop computer / laptop computer
Mac OS	Proprietary	
Ubuntu	Free and open source	
Fedora	Free and open source	
Android	Free and open source	Mobile phone
Windows Mobile OS	Proprietary	
iOS	Proprietary	
Windows Server	Proprietary	Computer network
Linux	Free and open source	

DO IT - PRACTICAL 1 (TEXTBOOK PAGES 138 - 141)

See the booting process and the software of your computer. **Ask your teacher first.**

1	Switch on a computer. The first text you see comes from the **firmware**.
2	When the desktop appears, the OS is in RAM and has full control.
3	Open the **Start** menu. Write the names of three application software.
4	Find the name of the OS: **Settings > System > About**. Is it proprietary?

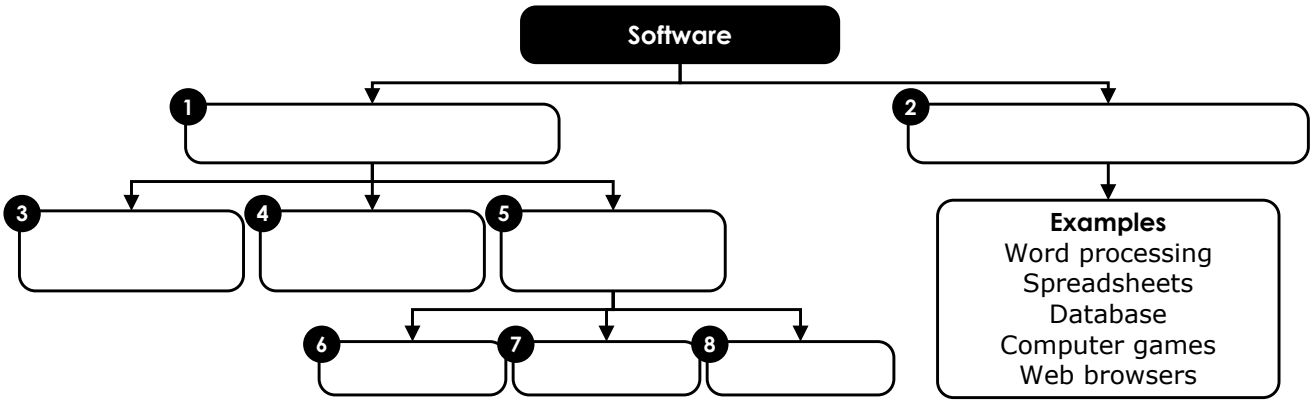
A Complete the software tree

8 marks

Write the words in the numbered boxes. Two words are not used.

WORD BANK

Compiler	Hardware	Operating system	Application software	Interpreter
Language translators	Firmware	System software	Utility software	Assembler



B Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

hardware	firmware	liveware	POST
Bootstrap	RAM	machine	translators

1. Any physical component of a computer system that can be touched is called ..... .
2. The booting instructions stored in the ROM are called ..... .
3. The users who operate the computer are called ..... .
4. When the computer starts, the BIOS performs the ..... (Power-On Self-Test).
5. The ..... Loader is located in the boot sector of the secondary storage.
6. The operating system is loaded from the secondary storage into ..... .
7. Language translators translate high-level languages into ..... language.
8. Assembler, compiler and interpreter are language ..... .

C Put the booting steps in order

5 marks

Write 1, 2, 3, 4 and 5 in the boxes.

a.		The Bootstrap Loader loads the operating system from the secondary storage into RAM.
b.		Electrical power is supplied to the processor, RAM and ROM.
c.		Full control of the computer is transferred to the operating system.
d.		The BIOS stored in ROM performs the POST (Power-On Self-Test).
e.		The BIOS loads and executes the Bootstrap Loader from the boot sector.

D Answer these

4 marks

Write full sentences.

1. What is firmware? (2 marks)

2. Write two things that the operating system does. (2 marks)

E Which part of the computer system?

8 marks

Tick one box in each row.

It is ...	Hardware	Firmware	Software	Liveware
The display unit, the keyboard and the mouse	☐	☐	☐	☐
The booting instructions stored in the ROM	☐	☐	☐	☐
Nimal, who operates the library computer	☐	☐	☐	☐
A computer game (a set of instructions)	☐	☐	☐	☐
The hard disk, the speaker and the printer	☐	☐	☐	☐
The first text on the screen at start-up	☐	☐	☐	☐
A word processing program	☐	☐	☐	☐
The users who operate the computer	☐	☐	☐	☐

CHAPTER 2

Types, Services and User Interfaces

? TYPE IT OR CLICK IT?

"My grandfather typed commands on his first computer. I just click on icons!"

"He used a command line interface. You use a graphical user interface."

Operating systems are of different **types** and give three main **services**.

5.6 Types of operating systems

The main function of an OS is to provide an environment suitable for executing the commands issued by the user. Based on what they do, operating systems are classified like this:

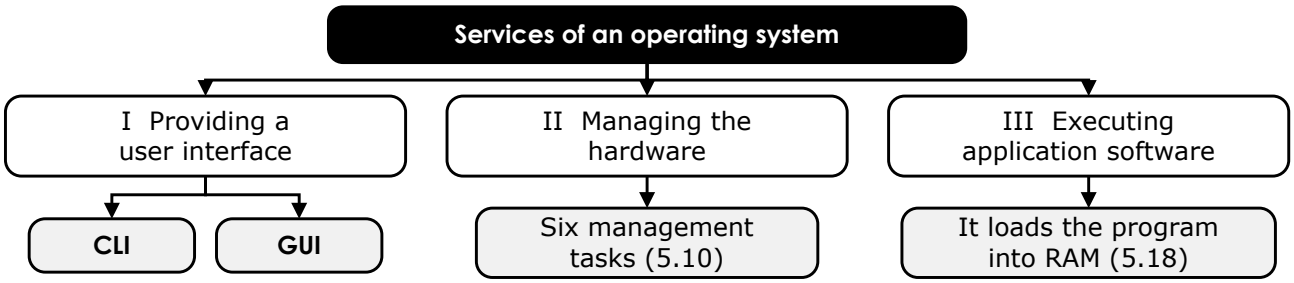
Type	What it does	Examples
Single user OS	Gives service to one person at a time.	MS DOS
Multi-user OS	Lets many users use one system at the same time, as in mainframe and server computers.	Linux, Windows Server
Multi-tasking OS	Runs many processes at the same time: one user can do many tasks at once.	Windows 7, Windows 8, Ubuntu, Mac OS
Real time OS	Gives the output without any observable delay. It is made for particular devices: automatically controlled machines, scientific devices and small gadgets.	ATM machines, calculators

"You do not learn ICT by reading. You learn it by doing."

- THE KING OF ICT

5.7 Services of an operating system

The OS manages the hardware and the other software in a computer system, and provides services to other software. It has **three main services**:



5.8 The command line interface (CLI)

We need a **user interface** to interact with the computer. There are two types: the **CLI** (Command Line Interface) and the **GUI** (Graphical User Interface).

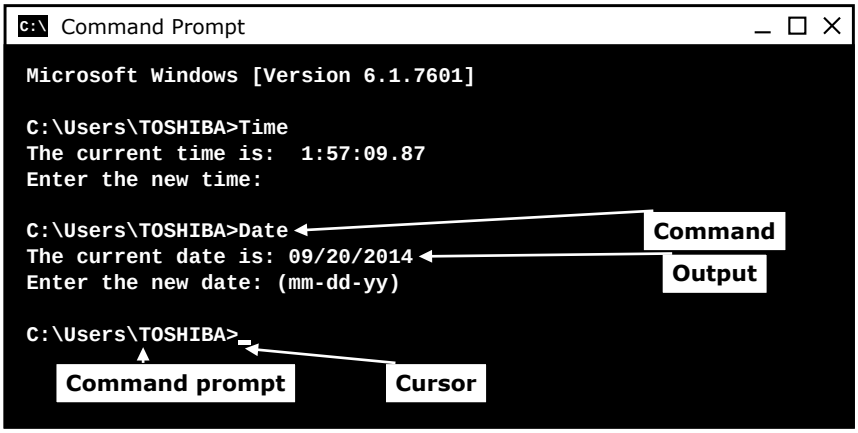


Figure 5.3 - a command line interface

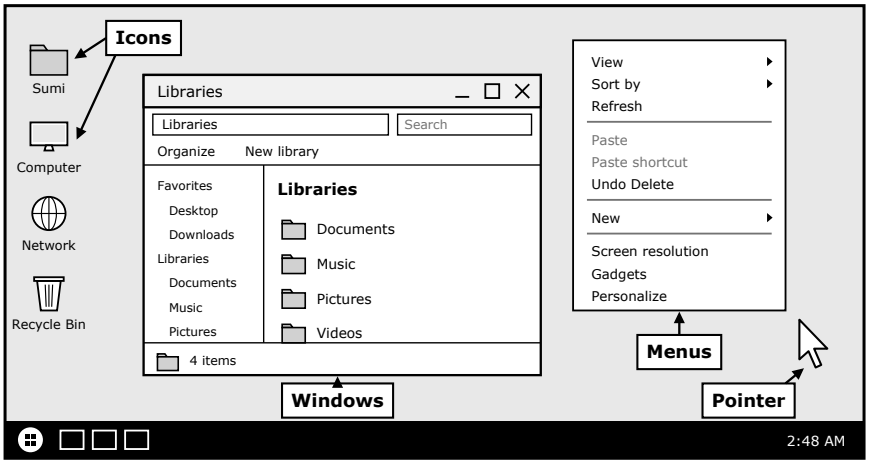
The CLI was used by early operating systems. It has a **prompt**, where the commands are keyed in using a keyboard.

The correct **syntax** has to be used: each command must be written exactly right.

Here the commands **Time** and **Date** were typed, and the computer showed the **output**.

5.9 The graphical user interface (GUI)

Almost all present-day operating systems use a **GUI**: we use the **mouse** or our **finger tips**, so computers are much easier to use. Its four components are **WIMP**:



W Windows

I Icons

M Menus

P Pointer

Figure 5.4 - a graphical user interface

DO IT - PRACTICAL 2 (TEXTBOOK ACTIVITY, PAGE 156)

Use a CLI and a GUI (textbook questions 4 - 8). **Ask your teacher first.**

1	Click **Start**, type **cmd** and press **Enter**. The Command Prompt (a CLI) opens.
2	Type **date** and press Enter twice. Then do the same with **time**.
3	Try **dir** (lists the files) and **cls** (clears the screen). Then type **exit**.
4	On the desktop, point to a window, an icon, a menu and the pointer. Name your desktop icons.

A Which type of operating system?

8 marks

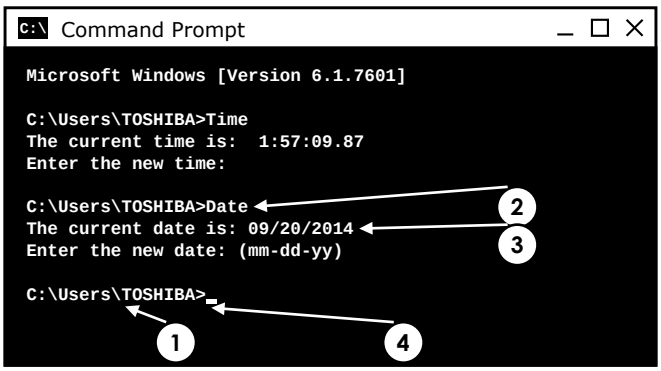
Tick one box in each row.

It shows ...	Single user	Multi-user	Multi-tasking	Real time
MS DOS serves one person at a time.	☐	☐	☐	☐
An ATM machine gives the output at once.	☐	☐	☐	☐
Many users share one mainframe computer.	☐	☐	☐	☐
Sunil types while music plays on Windows 8.	☐	☐	☐	☐
A calculator shows the answer at once.	☐	☐	☐	☐
Many users use Windows Server together.	☐	☐	☐	☐
Mac OS runs many programs at the same time.	☐	☐	☐	☐
Only one person at a time can use the OS.	☐	☐	☐	☐

B Label the screens

8 marks

Write the name of each numbered part.





1. ..... 2. ..... 3. ..... 4. .....

5. ..... 6. ..... 7. ..... 8. .....

"Some people type commands and some people click. Smart students learn both."

- THE KING OF ICT

C Match them

6 marks

Write the correct letter in the box.

A - Name		B - What it is
1. Single user OS		**A.** Runs many processes at the same time.
2. Multi-user OS		**B.** Uses windows, icons, menus and a pointer.
3. Multi-tasking OS		**C.** Gives service to one person at a time.
4. Real time OS		**D.** Gives the output without observable delays.
5. CLI		**E.** Commands are keyed in at a prompt.
6. GUI		**F.** Many users use the system at the same time.

D Choose the correct answer

4 marks

Underline your answer.

1. Which is NOT one of the three main services of an OS?
   (a) a user interface (b) managing the hardware
   (c) executing application software (d) typing documents

2. In Figure 5.3, C:\Users\TOSHIBA> is the
   (a) cursor (b) command prompt
   (c) output (d) command

3. Which is a multi-user operating system?
   (a) MS DOS (b) Windows Server (c) a calculator (d) an ATM

4. WIMP stands for
   (a) Windows, Icons, Menus, Pointer
   (b) Words, Images, Mouse, Printer
   (c) Windows, Internet, Memory, Programs
   (d) Web, Icons, Menus, Prompt

E True or false

6 marks

Write T or F in the box.

1.		The operating system manages the hardware and the other software of a computer.
2.		Executing application software is one of the three main services of an OS.
3.		The command line interface was used by early operating systems.
4.		In WIMP, the letter M stands for Mouse.
5.		A multi-tasking operating system can run only one task at a time.
6.		MS DOS is a multi-user operating system.

"Multi-tasking is great for computers. For homework, finish one task well at a time."

- THE KING OF ICT

CHAPTER 3

Managing the Hardware and Files

? WHERE IS MY PROJECT?

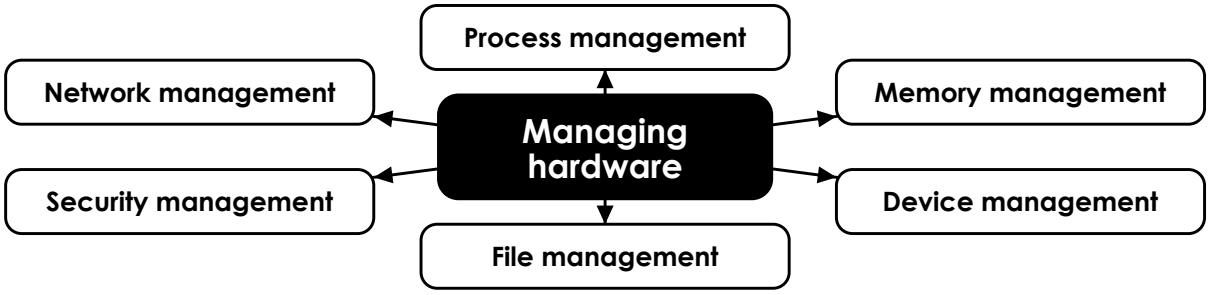
"I saved my science project yesterday. Now I cannot find it!"

"The OS keeps the location, the size and the date of every file. Look at its Properties!"

Managing the hardware has six tasks. **File management** is one of them.

5.10 Six management tasks

The hardware of a computer is managed by these six processes:



i PROCESS MANAGEMENT

A running computer program, or a part of it, is called a **process**. All the activities in a computer run as one process or as many processes. The OS allocates **CPU time**, **memory** and **input and output devices** to each process, and puts the processes in order (a sequence).

5.11 Memory management

Memory here means mainly the **RAM** (Random Access Memory). The OS uses the RAM and the secondary memory efficiently for the processes. The input data and the information wait in RAM:

1 INPUT	2 RAM	3 CPU	4 RAM	5 OUTPUT
Data comes from the input devices. →	The input data is stored in RAM. →	The CPU processes the data. →	The information is stored in RAM. →	It is sent to the output devices.

Memory management makes sure that **enough memory** is allocated for each process, and that the memory is **freed** when the process ends.

"A good manager shares time, memory and work fairly. So does a good class leader."

- THE KING OF ICT

5.12 Device management

Many peripheral devices are connected to a computer, and the OS manages them. **Device controllers** are used to control the peripheral devices, and **device drivers** are the software components used with them. Today many devices work as soon as they are connected: **plug and play**. The device driver is installed automatically the first time the device is connected.

5.13 File names and extensions

Every file name has two parts: the **file name** and the **file extension**.

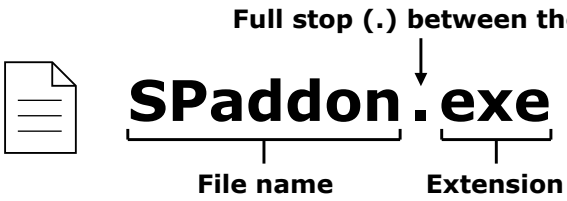


Figure 5.5 - the parts of a file name

Extension	File type
exe	Executable
docx	Word
xlsx	Excel
pptx	PowerPoint
accdb	Access

Note: the operating system identifies the **file type** using the file extension.

5.14 File operations and file properties

File operations of the OS: making new files and saving them, deleting unnecessary files, arranging and deleting folders, renaming files and folders, changing their storage location and creating backups.

File properties: a file is saved with a name and a location, but the OS keeps much more information about it. To see it, **right-click** the file and click **Properties**.

1. Location where the file is saved
2. Size of the file
3. The memory allocated by the storage to save the file
4. The date and time when the file was saved

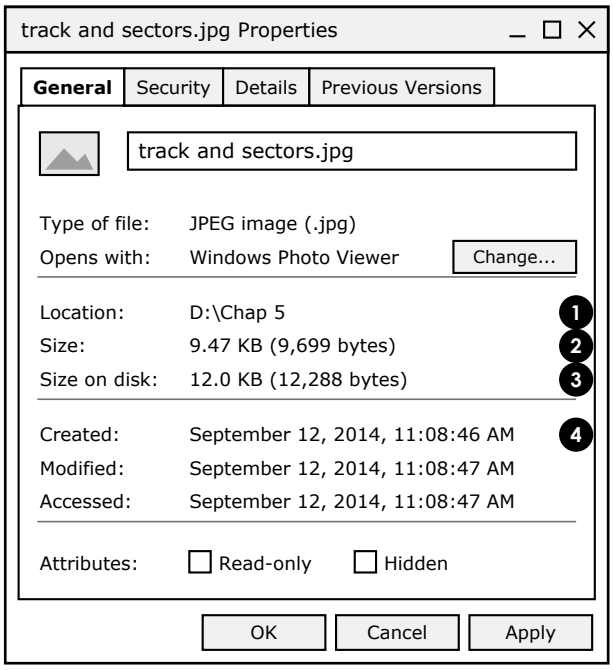
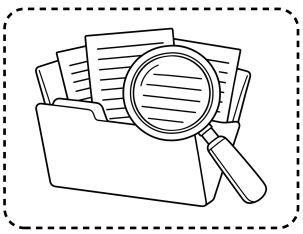


Figure 5.6 - viewing the details of a file

> "Give every file a good name and a good place, and you will always find it again."
>
> - THE KING OF ICT



Colour me!

5.15 File access methods and permissions

In **sequential access**, data in a file is read and written in order, from the beginning to the place you want: direct access is not possible. In **random access**, you can go directly to the place of the data you need.

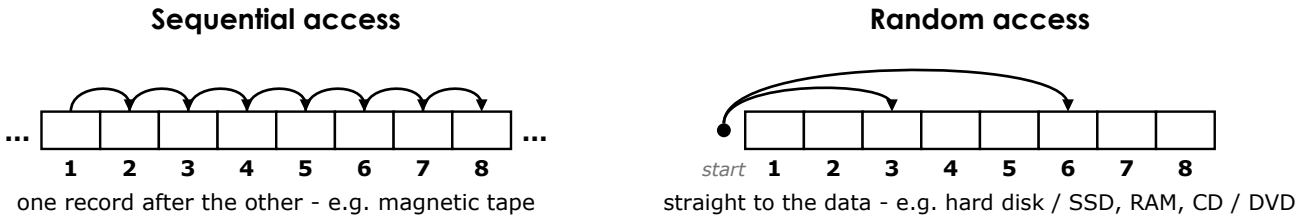


Figure 5.7 - sequential access and random access

File access permissions: the OS uses them to control which users can access a file, to keep the file secure. Reading, modifying (writing) and executing the file are allowed according to the privileges of each user.

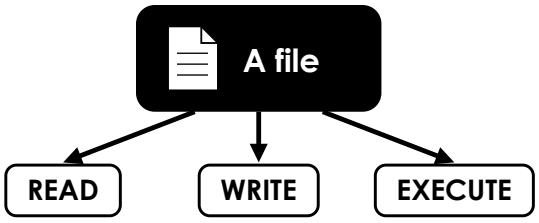


Figure 5.8 - file access permissions

5.16 File systems

A **file system** is the method used to organise, store and manage the files in a storage device:

File system	Maximum capacity of a file	File security	Operating systems
FAT16	**2 GB**	No	MS-DOS, Windows 95 / 98, Linux
FAT32	**4 GB**	**No**	Windows ME / 98, Linux, Mac OS
NTFS	**16 TB**	**Yes**	Windows XP / 7 / 8 / 10, Mac OS, Linux (the default for Windows)

5.17 Security and network management

! SECURITY MANAGEMENT

Threats: malware, such as a virus, and unauthorised access that deletes or destroys data. The OS protects the computer up to some extent with:

- password / PIN and biometric techniques to identify the users
- data encryption and file permissions
- third-party software: antivirus software and firewalls

i NETWORK MANAGEMENT

In a computer network, the OS:

- performs configuration, fault, performance and security management
- supports different types of network protocols
- supports wired and wireless connections for the devices
- helps to access one computer from a remote computer

5.18 Executing application software

This is the third main service of the OS. All application software is installed in the secondary memory. When we run a program:

1 INSTALLED	2 LOADED	3 IN CONTROL	4 FINISHED
The application is installed in the secondary memory.	When we run it, the OS loads it into RAM.	Control of the computer is handed over to the application.	When it ends, control returns to the OS.

DO IT - PRACTICAL 3 (TEXTBOOK ACTIVITY, PAGES 156 - 157)

Processes, devices, security and files (questions 1 - 3, 10, 12). **Ask your teacher first.**

1	Open the **Task Manager** (**Ctrl + Shift + Esc**). Open Notepad: it is a new process.
2	Connect a pen drive. Its device driver is installed automatically (plug and play).
3	Lock the computer (**Windows key + L**) and sign in. Find the antivirus and the firewall.
4	In File Explorer, tick **View > File name extensions**. Make a folder and a Word file in it.
5	Right-click the file > **Properties**: write its type, location, size and size on disk. Copy the folder to the pen drive (a **backup**).

A Which management task?

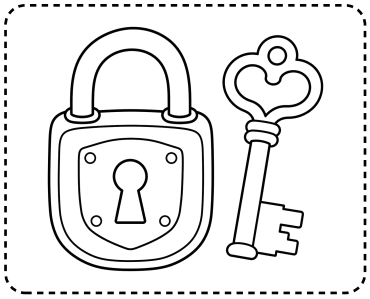
8 marks

Write the management task (5.10). Some are used twice.

What the OS does	Management task
The OS gives CPU time to each running program.	
The memory is freed when a process ends.	
A new printer works as soon as it is connected.	
The OS asks for a password or a PIN.	
The OS renames a file and makes a backup of it.	
A computer prints through the wireless network.	
The running processes are put in order.	
Antivirus software and a firewall protect the computer.	

"Never share your password. It is the key to your digital home."

- THE KING OF ICT

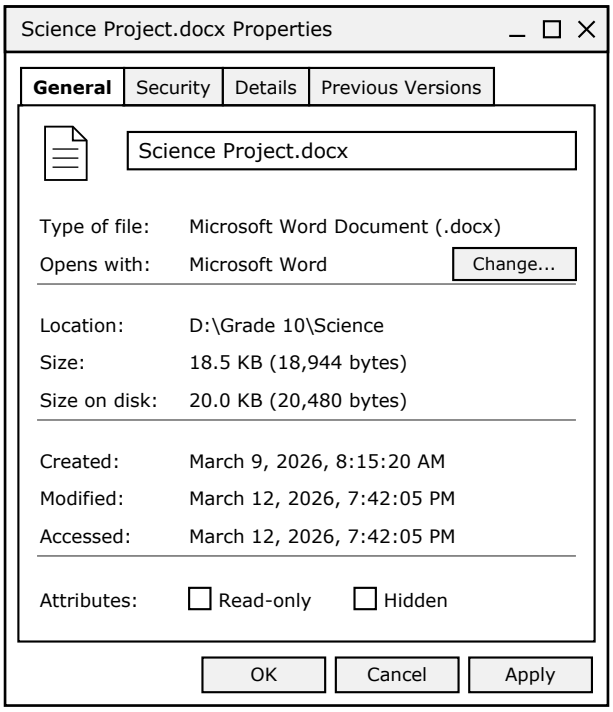


Colour me!

B Read the Properties window

6 marks

Look at the window. Answer the questions.



1. What type of file is it?
2. Which program opens it?
3. Where is it saved (its location)?
4. What is the size of the file?
5. How much memory did the storage allocate to save it?
6. When was the file created?

C Sequential or random access?

6 marks

Tick one box in each row.

Which access method?	Sequential	Random
Magnetic tape	☐	☐
A hard disk	☐	☐
RAM	☐	☐
A DVD	☐	☐
Reading records 1, 2, 3 ... in order to reach record 50	☐	☐
Going straight to record 50	☐	☐

D Complete the file systems table

5 marks

Write the missing values (textbook p. 148).

File system	Maximum capacity of a file	File security	Operating systems
FAT16		No	MS-DOS, Windows 95 / 98, Linux
FAT32			Windows ME / 98, Linux, Mac OS
NTFS			Windows XP / 7 / 8 / 10, Mac OS, Linux (the default for Windows)

> "A messy desk hides your books. A messy folder hides your files."
>
> - THE KING OF ICT

E Read the file names

10 marks

Write the extension and the file type of each file.

File	Extension	File type
Nimal.docx		
marks.xlsx		
setup.exe		
Science Day.pptx		
library.accdb		

CHAPTER 4

Utility Software, Drives and Disks

? MY COMPUTER IS SO SLOW!

"Our old computer takes so long to open my files. Is it broken?"

"Maybe its files are in pieces all over the disk. Defragmentation can help!"

Utility software helps the OS. You also meet **drives** and **disks**.

5.19 Utility software

Many **utility programs** in the OS help the machine to work and protect it from security threats. In the past we had to buy and install them; nowadays most of them come with the OS.

Utility software	Usage
Backup software	To copy files and take a backup of hard drives
Disk scanner	To check the errors in a hard drive
Task manager	To show the processes and programs in the computer and its general status
Anti-virus software	To protect the computer by identifying and removing malicious software
Screen saver	To blank the screen, or fill it with moving images or patterns, when it is not in use
Clipboard	Temporary storage of data / files for cut, copy and paste
File / data compression	To compress larger files into smaller files
Disk partitioning software	To divide one drive into several logical drives
Disk formatting	To prepare a partitioned disk for storing data, using file formats
Disk defragmentation	To organise the hard disk by rearranging clusters of small spaces together and creating a larger free space

5.20 Drives and folders

A computer saves data on the hard drive. If it is not partitioned, it is the **C:** drive; if it is partitioned, the partitions are **C:**, **D:**, **E:** ... in order. CD, DVD and Blu-ray drives get the next letters: with four partitions (C: to F:), a DVD drive is **G:**.

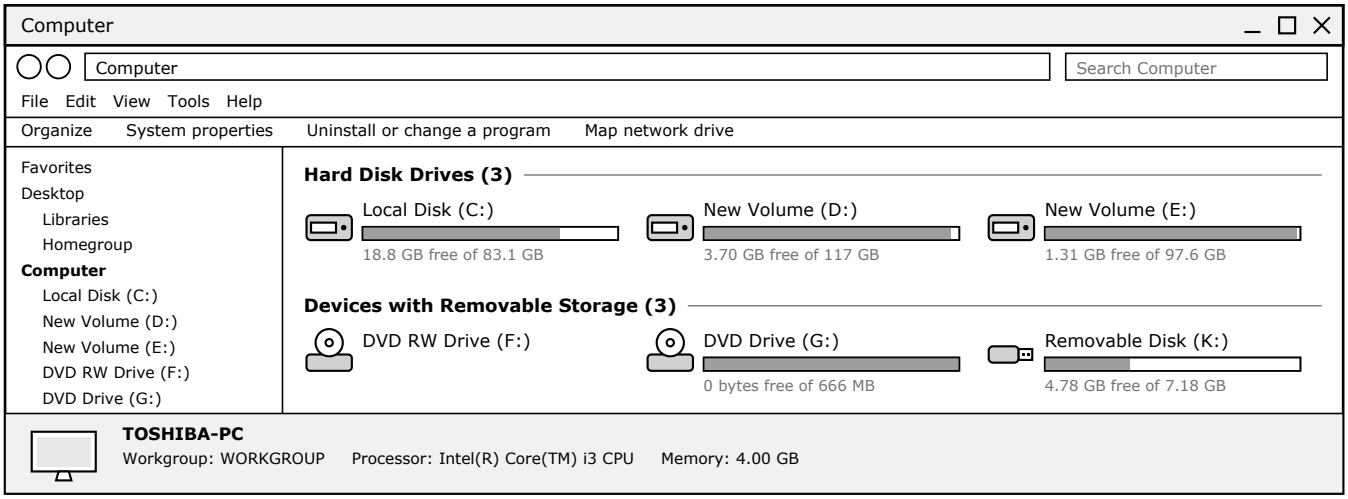
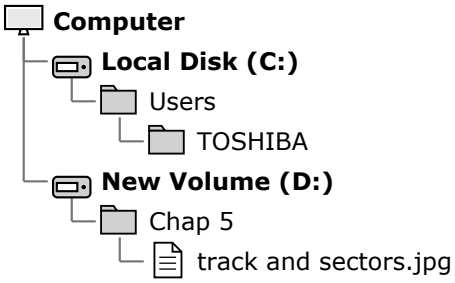


Figure 5.9 - viewing the drives: open Computer (This PC in Windows 10)

Pen drives and CD, DVD and Blu-ray drives are **devices with removable storage**. In Figure 5.9 the pen drive is K:. **Linux** has no drive letters: its drives are shown as folders, called **directories**.



Files are kept in **folders** on a drive. The **location** of a file shows the drive and its folders, joined by \.

The file **track and sectors.jpg** (Figure 5.6) is in the folder Chap 5 on drive D:, so its location is **D:\Chap 5**.

5.21 Disk partitioning

A hard drive has one drive at first. Dividing one drive into several logical drives is **disk partitioning**. It is normally done when a hard disk is configured for the first time, added or replaced. We need partitioning:

- to save different items in different places (software in one drive, other data in another)
- to have more than one OS on the same machine (for example Windows and Ubuntu)
- to meet the needs of the OS (sometimes the OS must have a separate partition)

! BACKUP FIRST!

When a partitioned hard disk is partitioned again, **all the data in it is deleted**. Keep a backup copy of the data first.

5.22 Disk formatting

After partitioning, each drive must be **formatted**. Formatting prepares a storage device (hard disk, SSD, floppy disk or USB flash drive) for storing data, using a file format in the OS. USB flash drives come pre-formatted. Formatting deletes the data, so keep a backup first. Usually a hard disk is partitioned and formatted before an OS is installed for the first time.

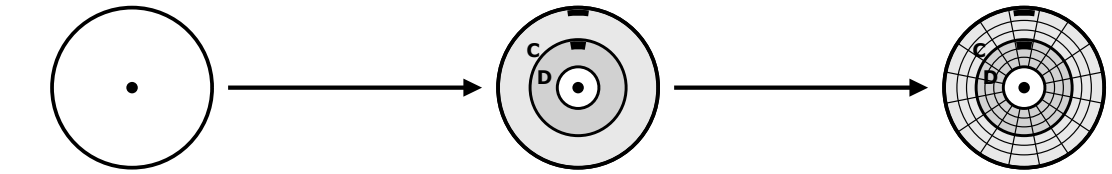


Figure 5.11 - a hard disk before and after partitioning and formatting

5.23 Defragmentation

Sometimes a large file is broken into pieces stored in different parts of the disk: this is **fragmentation**. It happens when a saved file grows after many edits, and when the spaces of deleted files are used for new files.

A fragmented file takes longer to read, so the computer slows down. **Defragmentation** rearranges the disk so that the pieces come together.

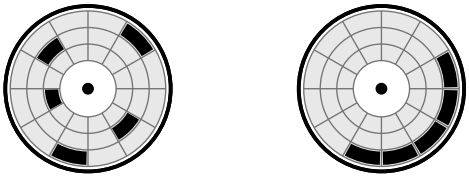


Figure 5.13 - defragmentation

(i) LINUX

Linux leaves ample space between saved files and can move a file that grows too big, so fragmentation rarely happens. Linux needs no defragmentation utility.

(>) DO IT - PRACTICAL 4 (TEXTBOOK ACTIVITY, PAGE 157)

Drives and utility programs (questions 13 - 15). Never format or partition a drive.

1	Open **This PC** (Computer). Write the letters of the hard disk drives and the removable devices.
2	Connect a pen drive: which letter does it get? Right-click drive C: and click **Properties**.
3	With your teacher, open **Defragment and Optimize Drives**. Click **Analyze** only.
4	Name five utility programs on your computer and say what they do (question 13).

A Read the drives window

8 marks

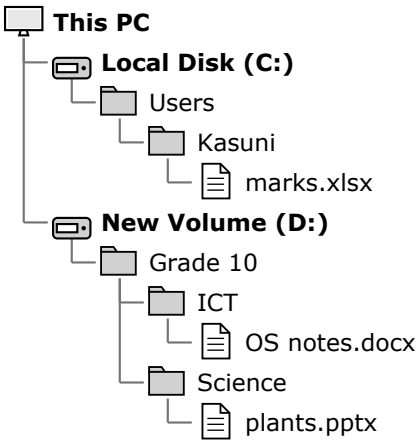
Use Figure 5.9 in 5.20 for questions 1 - 6.

1. How many hard disk drives does this computer have?
2. Which drive has the most free space?
3. What is the total size of drive D:?
4. What is the letter of the pen drive (Removable Disk)?
5. How much space is used on the pen drive?
6. Which drives are devices with removable storage?
7. Sunil's hard disk has two partitions, and his computer has one DVD drive. Write the drive letters of the partitions and the DVD drive. (2 marks)

B Where is the file?

4 marks

Look at the folders of Kasuni's computer.



1. Write the location of OS notes.docx.
2. Write the location of plants.pptx.
3. Write the location of marks.xlsx.
4. Which file is a PowerPoint file?

C Which utility?

8 marks

Write the name of the utility software (the table in 5.19).

What is needed	Utility software
Kamal wants a copy of his files in case the hard drive fails.	
Moving patterns fill the screen when nobody uses it.	
Nimali wants to see the running programs.	
A large file must be made into a smaller file.	
A copied sentence is kept until you paste it.	
The errors in the hard drive must be found.	
The computer must be protected from a virus.	
Small free spaces must be joined into a larger space.	

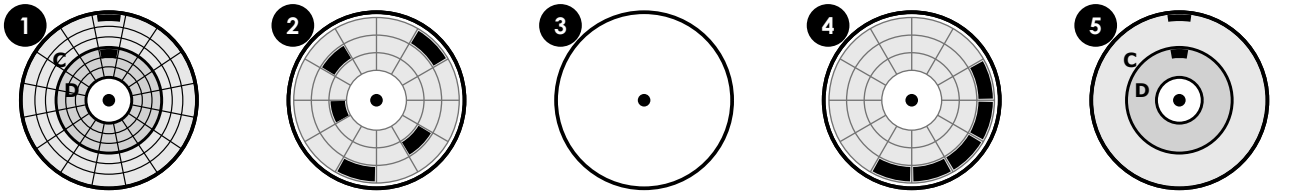
D Name the disks

5 marks

Write what each disk shows. Use the words in the box.

WORD BANK

before partitioning | after partitioning | after formatting | fragmented | defragmented



E True or false

8 marks

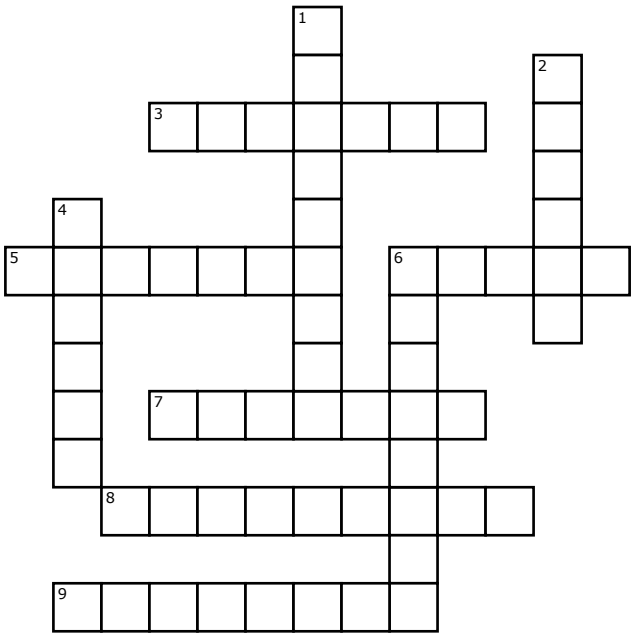
Write T or F in the box.

1.		In the past, utility programs had to be bought and installed separately.
2.		If a hard drive is not partitioned, it is labelled as the D: drive.
3.		In Linux, the partitions and flash drives are shown as directories (folders).
4.		When a hard disk is partitioned again, all the data on it is deleted.
5.		USB flash drives must always be formatted before they are used for the first time.
6.		Formatting a drive keeps all the data in it.
7.		A fragmented file takes longer to read.
8.		Linux based operating systems need a defragmentation utility.

F Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

3. The process that loads the OS when the computer starts (7)
5. A running computer program, or a part of it (7)
6. A free and open source OS with no drive letters (5)
7. The P in WIMP (7)
8. It keeps data for cut, copy and paste (9)
9. Booting instructions stored in the ROM (8)

DOWN

1. The part of a file name after the full stop (9)
2. Make this copy before you format a drive (6)
4. In a CLI, the commands are keyed in at the ... (6)
6. The users who operate the computer (8)

REVISION

Unit 5 in One Page

Read this the night before the test

Topic	The key facts
Computer system	Hardware (can be touched), firmware (booting instructions in ROM), software (sets of instructions) and liveware (the users).
Booting	Power on, the BIOS in ROM runs the POST, then loads the Bootstrap Loader, which loads the OS into RAM.
Software	System software: the OS, utilities, translators. Application software: word processing, spreadsheets, database, games, browsers.
Operating systems	Proprietary: Windows, Mac OS, iOS, Windows Server. Free and open source: Ubuntu, Fedora, Android, Linux.
Types of OS	Single user (MS DOS), multi-user (Windows Server), multi-tasking (Windows 8, Mac OS), real time (ATM, calculator).
Three services	Providing a user interface, managing the hardware, executing application software.
CLI and GUI	CLI: commands keyed in at a prompt, correct syntax. GUI: WIMP - Windows, Icons, Menus, Pointer; the mouse or finger tips.
Managing hardware	Process, memory, device, file, security and network management.
Processes and memory	A process is a running program. Input data and information wait in RAM. Plug and play installs the device driver automatically.
Security	Password / PIN, biometrics, encryption, file permissions, antivirus software and firewalls.
Files	File name . extension (exe, docx, xlsx, pptx, accdb). Properties: type, location, size, size on disk, dates.
Access and file systems	Sequential (magnetic tape) or random (hard disk, SSD, RAM, CD / DVD). FAT16 2 GB, FAT32 4 GB, NTFS 16 TB with file security.
Utilities	Backup, disk scanner, task manager, antivirus, screen saver, clipboard, compression, partitioning, formatting, defragmentation.
Drives and disks	Partitions C:, D:, E: ... (directories in Linux). Partition, then format. Defragmentation brings the pieces of files together.

TICK WHEN YOU CAN DO IT

- [ ] I can explain hardware, firmware, software, liveware and the booting process.
- [ ] I can sort software into system software and application software.
- [ ] I can name the types of operating systems and give an example of each.
- [ ] I can label a CLI screen and the WIMP parts of a GUI.
- [ ] I can explain the six management tasks of an operating system.
- [ ] I can read a file name, its extension and its Properties window.
- [ ] I can explain utility software, drives, partitioning, formatting and defragmentation.

WORD POWER

Unit 5 Word List

Twenty-six words you must know

Word	What it means
Hardware	Any physical component of a computer system that can be touched.
Firmware	The booting instructions stored in the ROM.
Software	A set of instructions given to the computer to perform some activity.
Liveware	The users who operate the computer.
BIOS	Basic Input Output System: the booting commands stored in ROM (textbook page 51).
POST	Power-On Self-Test. The BIOS checks the hardware and sets up the computer.
Bootstrap Loader	It is in the boot sector and loads the OS into RAM.
Booting	The process that starts the computer and gives full control to the OS.
System software	The operating system, utility software and language translators.
Application software	Software for the user's work: word processing, games, web browsers ...
Operating system	Software that manages the hardware and the software of a computer.
Language translator	Translates high-level languages into machine language (0s and 1s).
Multi-tasking OS	An OS that runs many processes (tasks) at the same time.
Real time OS	An OS that gives the output without observable delays (ATM machines).
CLI / GUI	Command line interface (typed commands) / graphical user interface (WIMP).
Process	A running computer program, or a part of it.
Device driver	Software for a device. Plug and play installs it automatically.
Malware	Malicious software, such as a virus.
File extension	The part of a file name after the full stop, such as docx.
File system	The method used to organise, store and manage files: FAT16, FAT32, NTFS.
Random access	Going directly to the place of the data, as in a hard disk.
Utility software	Programs that help the computer to work well, such as backup software.
Partitioning	Dividing one drive into several logical drives.
Formatting	Preparing a storage device for storing data.
Fragmentation	A file broken into pieces in different parts of the disk.
Defragmentation	Rearranging the disk so the small spaces join into a larger free space.

"Learn the words today, and the ideas will follow tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 5 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. The users who operate the computer are called
   (a) hardware (b) firmware (c) liveware (d) software
2. Which of these is a free and open source operating system?
   (a) iOS (b) Mac OS (c) Ubuntu (d) Windows Server
3. Which of these is application software?
   (a) a compiler (b) a disk scanner (c) a web browser (d) an interpreter
4. Which operating system does the textbook give for computer networks?
   (a) Windows Server (b) iOS (c) Android (d) MS DOS
5. In WIMP, the letter I stands for
   (a) Internet (b) Icons (c) Input (d) Interface
6. Which type of file is report.xlsx?
   (a) Word (b) Excel (c) PowerPoint (d) Access
7. Which of these uses sequential access?
   (a) magnetic tape (b) an SSD (c) a hard disk (d) a DVD
8. A - USB flash drives come pre-formatted. B - Linux needs a defragmentation utility. C - Linux shows its drives as directories. Which are correct?
   (a) A and B only (b) B and C only (c) A and C only (d) A, B and C

2 Fill in the blanks

8 marks

One word in each blank.

1. An OS provides an ..... suitable for executing the user's commands.
2. Input data is stored in the ..... before it goes to the CPU.
3. Assembler, compiler and interpreter are language ..... .
4. An ATM machine uses a ..... time operating system.
5. In a command line interface, the correct ..... has to be used.
6. Plug and ..... devices work as soon as they are connected.
7. The ..... keeps cut or copied data for a short time.
8. A screen ..... blanks the screen when the computer is not in use.

3 Name the pictures

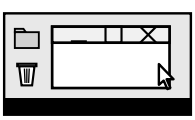
8 marks

Use the words in the box. Two words are not used.

WORD BANK

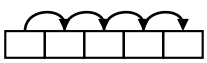
Random access	Firewall	CLI	Multi-user OS	Fragmented hard disk
Device driver	GUI	Sequential access	Partitioned hard disk	File access permissions

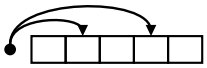
1 

2 

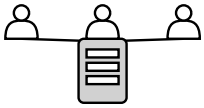
3 

4 

5 

6 

7 

8 

4 Answer these

16 marks

Write in the space given.

1. Write the four steps of the booting process in order. (4 marks)

   1. ..... 2. .....

   3. ..... 4. .....

2. Name the three main services of an operating system. (3 marks)

   I ..... II ..... III .....

3. Look at the file name Kamal.pptx. Write (a) its extension, (b) its file type and (c) what the OS uses to identify the type of a file. (3 marks)

   (a) ..... (b) ..... (c) .....

4. Write two differences between the FAT32 and NTFS file systems. (2 marks)

5. Why should you make a backup before you partition or format a disk? (2 marks)

6. What is a process? Write one thing that the OS allocates to each process. (2 marks)